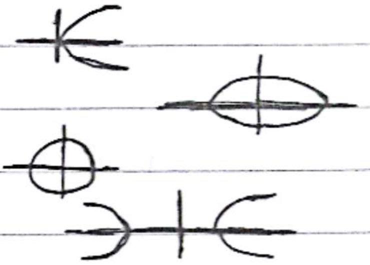
Conic Sections

Def:

A Conic Section is the intersection of a plane and a right circular cone. The four basic types of Conics are

1) Parabolas
2) ellipses
3) circles
4) hyperbolas



The equation of every Conic can be written in the following form

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$

Now

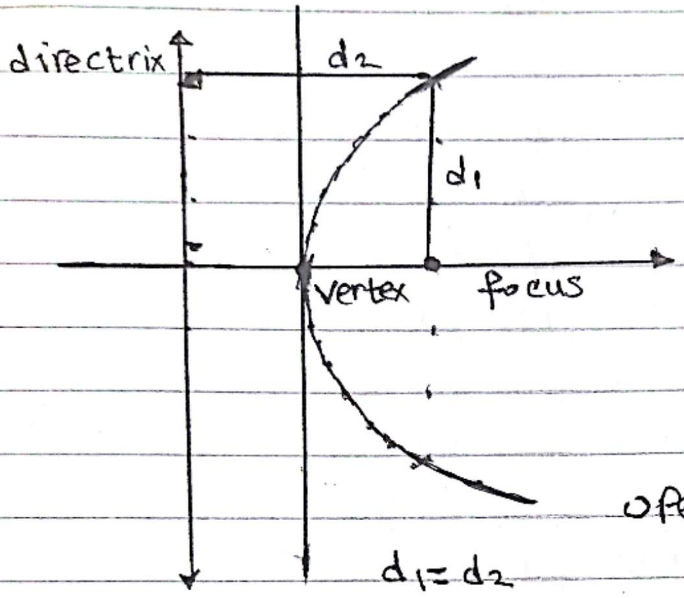
1). If $B^2-4AC<0$, the Conic is a circle if $A=C$ and $B=0$, or an ellipse

2) If $B^2-4AC=0$, the Conic is a parabola.

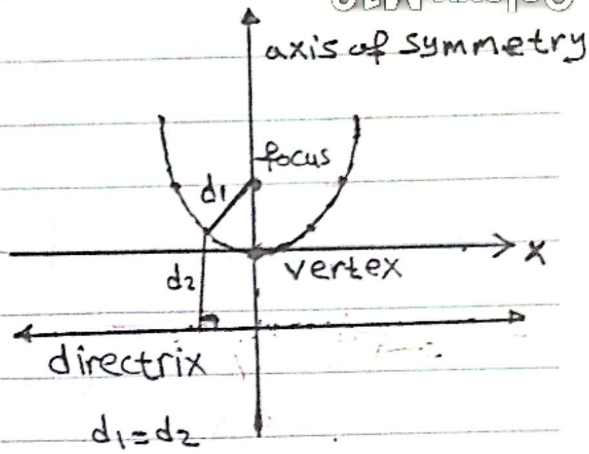
3) If $B^2-4AC>0$ " " " " hyperbola.

Parabola

a parabola defined as the set of all points in a plane that are the same distance from a line called (the directrix) and a point not on the line (directrix) called the (focus). The midpoint of line segment perpendicular to the directrix and passing through the focus is the (vertex) of the parabola. The line that passes through the vertex and the focus is the (axis of symmetry). The axis of symmetry is perpendicular to the directrix.



opens horizontally right



opens vertically up

Focal Chord: A chord of the parabola passing throug the focus is called a focal chord

Latus rectum A focal chord of a parabola perpendi-icular to its principle axis (symmetry) is called latus rectum

Focal width : the length of latus rectum

Standard Equations of a parabola with vertex (h,k)

Direction	opens vertically (up or down)	opens horizontally (left or right)
Equation	$(x-h)^2 = 4a(y-k)$	$(y-k)^2 = 4a(x-h)$
Focus	$(h, k+a)$	$(h+a, k)$
Axis of symmetry	$x=h$	$y=k$
Directrix	$y=k-a$	$x=h-a$
If $a>0$	opens up	opens right
If $a<0$	opens down	opens left

Theorem: The focal width of the parabola $= |4a|$.

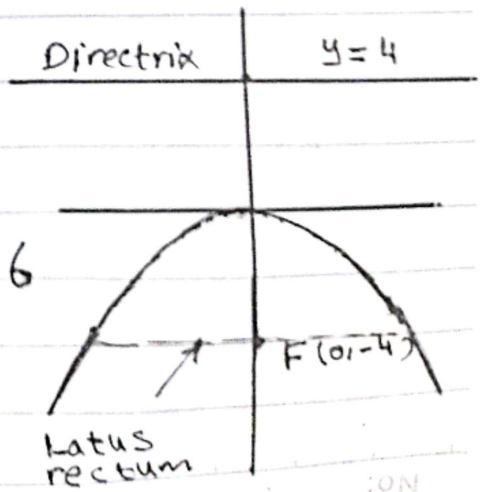
EX. I Graph $x^2 = -16y$ and identify the vertex, the directrix, focus and focal width.

Solution

$x^2 = -16y = 4(-4)y$ opens vertically, $a=-4<0$

$(x-h)^2 = 4a(y-k)$ $(h,k)=(0,0)$ (down

- vertex $(h,k)=(0,0)$
- directrix $y=-(-4)=4$
- focus $(0,-4)$
- focal width $|4a| = |4(-4)| = 16$



Ex.2 Graph $y^2=8x$ and identify the vertex, directrix, focus, and focal width.

Solution

Compare with the Standard Equation

$$(y-k)^2=4a(x-h)$$

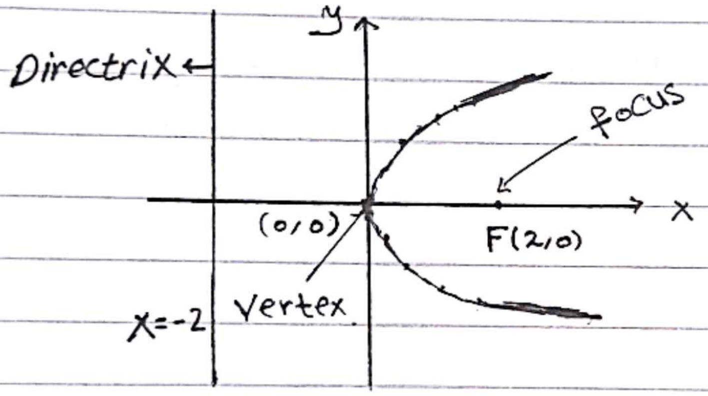
$$y^2=4(2)x \qquad a=2 \quad \text{(open horizontally- right)}$$

Vertex $(h,k)=(0,0)$

directrix $x=-a \Rightarrow x=-2$

focus $(h+a,k)=(2,0)$

focal wids $=|4a|=8$.



Ex.3

Find the equation of a parabola having the origin as its vertex, the y axis as its axis of symmetry and $(-10,-5)$ lies on it.

Solution

since vertex $=(0,0)$

axis of symmetry y-axis i.e $x=0$

$\Rightarrow$ The equation of Parabola

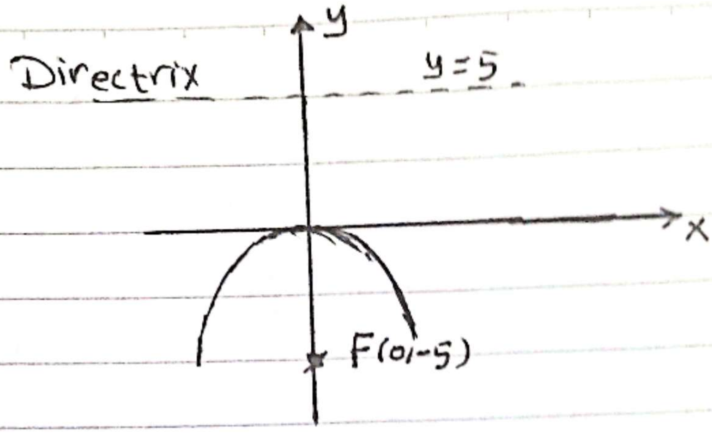
$$\therefore \quad x^2=4ay$$

$\because (-10,-5)\in$ Parabola $\Rightarrow (-10)^2=4a(-5) \Rightarrow a=-5$

$\Rightarrow$ The equation of parabola is $x^2=-20y$

$\Rightarrow$ Focus $= (0, -5)$

directrix $y = 5$



EX. 4. Find the equation of a parabola having the origin as its vertex, its focus is $(-4/3, 0)$ and the equation of the directrix is $x = 4/3$ and find the focal width.

solution

$(h, k) = (0, 0)$.

$(a, 0) = (-4/3, 0) =$ Fosus

$x = -a \qquad x = 4/3$ directrix

equation: $y^2 = 4ax$

$$y^2 = 4\left(\frac{-4}{3}\right)x = \frac{-16}{3}x$$

focal widt $= |4a| = \frac{16}{3}$.

EX. 5 Graph $3x^2 = 5y$ and identify the directrix, focus, and focal width

solution

$$3x^2 = 5y \Rightarrow x^2 = \frac{5}{3}y \Rightarrow \frac{5}{3} = 4a \Rightarrow a = \frac{5}{12}$$

then the equation represents a parabola with

- Vertex $(0, 0)$

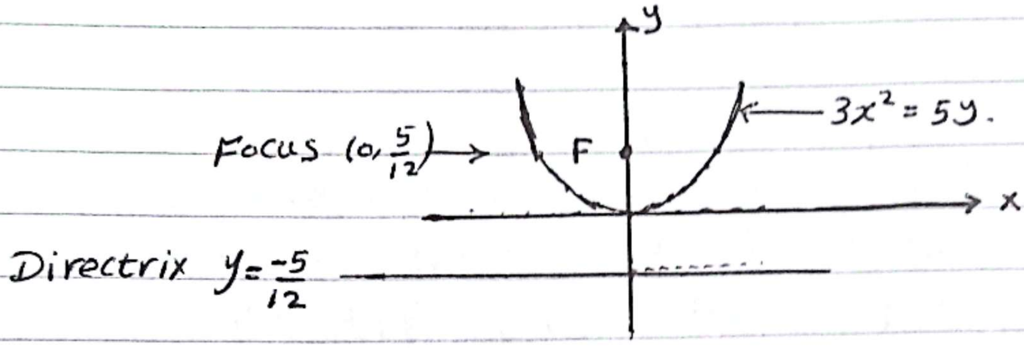
Axis of ymmetry $\quad x=0 \quad$ i.e y axis

Focus $\quad (0,a) = (0, \frac{5}{12})$

directix $\quad y=-a \Rightarrow y = -5/12$

focal width $\quad |4a| = 4(\frac{5}{12}) = \frac{5}{3}$

Direction $\quad$ open vertically up



Remark: (1) The general equation for a parabola symmetrized to x-axis is

$$x = Ay^2 + By + C$$

(2) The general equation for a parabola symmetrized to y-axis is

$$y = Ax^2 + Bx + C.$$

Ex. 6 $\quad$ Graph $y = \frac{1}{4}(x-2)^2 - 3$ and identify the directrix, focus, and focal width.

solution.

$y = \frac{1}{4}(x-2)^2 - 3 \Rightarrow (x-2)^2 = 4(y+3) \Rightarrow$

$\underline{(x-h)^2 = 4a(y-k)} \Rightarrow (h,k) = (2,-3), a=1$

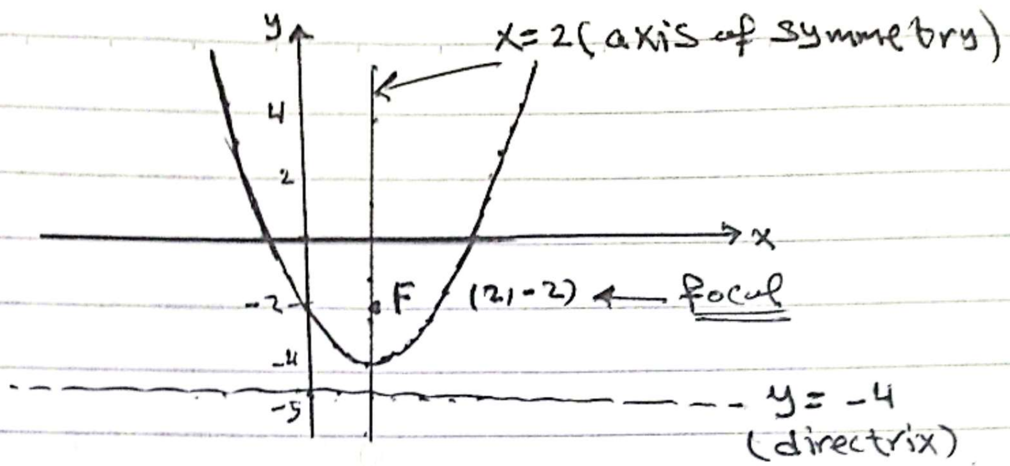
Direction : open vertically up $(a>0)$

Focus $\quad (h, k+a) = (2,-2)$

Axis of symmetry $\quad x = h \Rightarrow x = 2$

Directrix $\quad y = k - a \Rightarrow y = -4$

focal width $\quad |4a| = 4$



EX

Write the equation of the parabola with Vertex (-1,3), and a focus (-4,3), and identify the directrix, axis of symmetry and focal width

Solution

verex

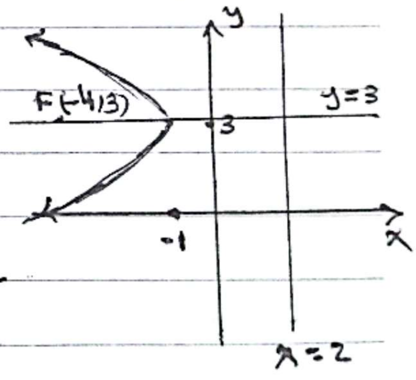
$(-1,3)$ (h,k) $h=-1$, $k=3$

Focus $\Rightarrow$ $\Rightarrow a=-3$

$(-4,3)$ $(h+a,k)$ $h+a=-4$

equation of the parabola $(y-k)^2 = 4a(x-h) \Rightarrow$

$(y-3)^2 = 4(-3)(x+1)$



- Directrix $x = h-a \Rightarrow x = -1+3 = 2$
- axis of symmetry $y = k \Rightarrow y = 3$
- focal width $|4a| = |4(-3)| = 12$

Ex.

Write the equation of the parabola with directrix $y=-1$ and a focus $(2,-5)$. Identify axis of symmetry and focal width.

Solution

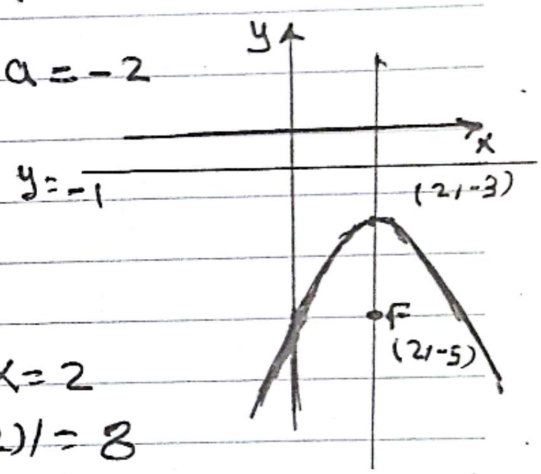
$$(x-h)^2 = 4a(y-k)$$

Directrix $\quad y = k-a \quad = -1$

focus $\quad (h, k+a) \quad = (2,-5)$

$\therefore h = 2, \quad k-a=-1$
$\quad\quad\quad\quad\quad k+a=-5 \Rightarrow 2k=-6 \Rightarrow k=-3$

$\therefore a = -2$



$\therefore$ The equation is

$$(x-2)^2 = 4(-2)(y+3)$$

axis of symmetry $\quad x=h \Rightarrow x=2$

focal width $\quad |4a| = |4(-2)| = 8$

Ex. Graph $y = -x^2+2x-6$ and identify the directrix, focus, and focal width.

Solution

By completing square, we obtain

$$y = -x^2+2x-6 = -(x^2-2x+1)-5 = -(x-1)^2-5$$

$$\Rightarrow (x-1)^2 = -(y+5) \quad \equiv \quad \underline{(x-h)^2 = 4a(y-k)}$$

vertex $\quad (1,-5) \quad 4a=-1 \Rightarrow a = -1/4$

focus $\quad (h,k+a) = (1,-5-1/4) = (1, \frac{-21}{4})$

focal width $\quad |4a| = 1$

directrix $\quad y = k-a \Rightarrow y = -5+1/4 = \frac{-19}{4}$

Ex.

Graph $y^2+4y-12x+40=0$ and identify, the directrix, focus, and focal width

Solution.

By completing squares we obtain

$$y^2+4y-12x+40 = (y^2+4y+4) - 12x+40-4 =$$

$$= (y+2)^2 - 12x+36 = (y+2)^2 - 12(x-3) = 0$$

$\Rightarrow$

$$(y+2)^2 = 12(x-3) \equiv \underline{(y-k)^2 = 4a(x-h)}$$

vertex $(h,k) = (3,-2)$, $4a=12 \Rightarrow a=3$

focus $(h+a,k) = (6,-2)$

Directrix $x = k - a \Rightarrow x = 3-3 = 0$

focal widt $|4a| = 12$.